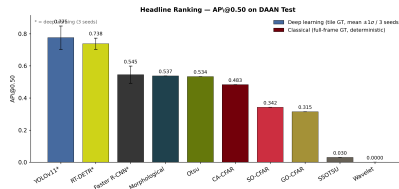

Classical vs. Deep Learning Object Detection in X-Band Marine Radar Imagery

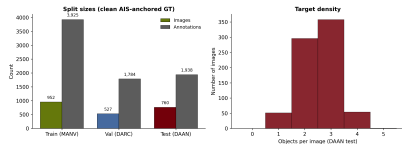
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First cross-paradigm benchmark of 7 classical + 4 deep detectors on the DLR DAAN / DARC / MANV corpus, under a single AIS-anchored ground truth with per-dataset polar calibration.



Why marine radar detection?

- X-band marine radar produces **plan-position-indicator (PPI)** imagery.
- Vessels, buoys, and debris appear as bright returns over sea clutter.
- Core problem for **maritime safety** and **autonomous surface vehicles**.
- Detection pipelines historically split into two camps:
 - Classical signal/image processing (CFAR, morphology, wavelets, Otsu).
 - Modern deep detectors (YOLO, DETR, Faster R-CNN, U-Net).



DLR corpus at a glance: per-sub-dataset split sizes and target-density distribution across the 2,239 evaluated frames.

The gap in the literature

- Recent DL results on marine radar look spectacular: YOLOv11 **mAP@0.5 > 0.95** [Heuer 2025]; Marine-Faster R-CNN **93.6%** recall [Chen 2022].
- Classical methods (CA/GO/SO/OS-CFAR, SWT, Otsu) have **decades** of theory but sparse modern reporting.
- **No single study** evaluates both paradigms on **identical data, identical GT, and identical metrics**.

Research question

On a shared X-band PPI benchmark, how large is the classical-vs-deep gap when every detector is evaluated under the same AIS-anchored ground truth and the same COCO protocol?

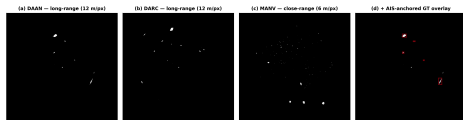
Eleven detectors, two paradigms

Family	Method	Core idea
Classical CFAR	CA-CFAR	Cell-averaged adaptive threshold
	GO-CFAR / SO-CFAR	Greatest-of / smallest-of leading & lagging cells
	OS-CFAR	Order-statistic (rank-based) threshold
Other classical	Morphological	Otsu → opening → connected components
	Wavelet (SWT)	2-D stationary wavelet high-freq. sub-band threshold
	SSOTSU	Seed-point Otsu for small-target segmentation
Deep learning	YOLOv11 (with SAHI)	1-stage anchor-free, tiled inference
	RT-DETR	Transformer end-to-end, NMS-free
	Faster R-CNN	2-stage anchor-optimised for radar
	U-Net	Per-pixel segmentation → bbox

All detectors receive identical preprocessed PPI input and emit COCO-format boxes for unified evaluation.

Dataset: DLR DAAN / DARC / MANV

- **DLR open marine radar corpus** — three sub-datasets:
 - **MANV** — harbour manoeuvres, **6 m/px**, close-range.
 - **DARC** — open water, **12 m/px**.
 - **DAAN** — open water, **12 m/px**.
- **2,239** PPI frames and **11,893** target instances total.
- Annotations include **AIS transponder track**, bounding boxes, and segmentation.
- Split (paper convention): MANV → train, DARC → val, DAAN → test.



Preprocessing: polar PPI → masked, calibrated imagery with hybrid AIS-anchored GT boxes.

Fixing the public DLR release

Two systematic issues in the released annotations had to be corrected **before** any detector comparison was meaningful.

1. Polar-origin bias (17 px)

Visual PPI centre and polar projection origin were offset by ~ 17 px. Naive AIS projection placed seeds **40–70 px** from their radar blobs.

Fix: separate visual-centre from polar-origin; per-dataset calibration reduces residuals to 5–7 px.

2. MANV range-scale error ($2\times$)

Published range-per-pixel disagreed with ground truth by a factor of two on MANV.

Fit a per-dataset range scale. Fix: dataset-aware calibration of range resolution before projecting AIS tracks into pixel space.

Hybrid AIS-anchored ground truth

AIS seeds in pixel space $\rightarrow$ image-grounded connected-component refinement $\rightarrow$ tight bounding boxes. Replaces DBSCAN 33×33 padded boxes used in prior work.

- **COCO suite**: AP@0.50, AP@0.75, AP@[.50:.95], AR@100.
- Deep detectors: **three random seeds** (42, 123, 456); report mean $\pm 1\sigma$.
- Classical detectors: deterministic; one run per method.
- Evaluation surfaces:
 - DL — 4,560 tiles of 640×640 px.
 - Classical — 760 full PPI frames of 1050×1024 px.
- Same AIS-anchored GT geometry on both surfaces.

Pending analyses (see limitations)

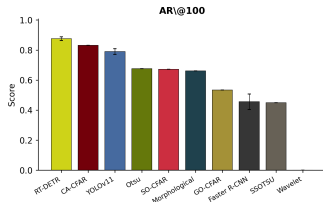
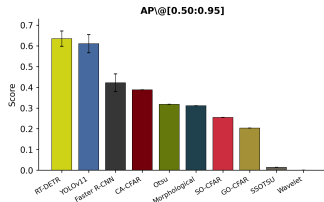
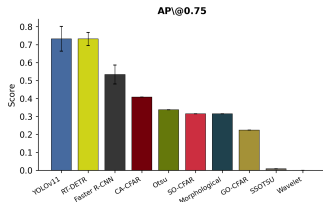
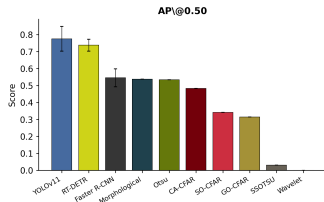
TIDE error decomposition;
pairwise bootstrap significance;
Pareto accuracy-vs-latency curves.

Main results: DAAN test

Method	Type	AP@.50	AP@.75	AP@[.50:.95]	AR@100
YOLOv11	DL ($\times 3$)	0.776 $\pm$.06	0.733 $\pm$.04	0.610 $\pm$.03	0.790 $\pm$.03
RT-DETR	DL ($\times 3$)	0.738 $\pm$.03	0.732 $\pm$.03	0.635 $\pm$.02	0.876 $\pm$.01
Faster R-CNN	DL ($\times 3$)	0.545 $\pm$.04	0.533 $\pm$.04	0.422 $\pm$.04	0.456 $\pm$.04
Morphological	Classical	0.537	0.315	0.311	0.661
Otsu	Classical	0.534	0.338	0.318	0.676
CA-CFAR	Classical	0.483	0.408	0.388	0.832
SO-CFAR	Classical	0.342	0.315	0.255	0.672
GO-CFAR	Classical	0.315	0.224	0.204	0.535
SSOTSU	Classical	0.030	0.009	0.014	0.449
Wavelet	Classical	0.000	0.000	0.000	0.000

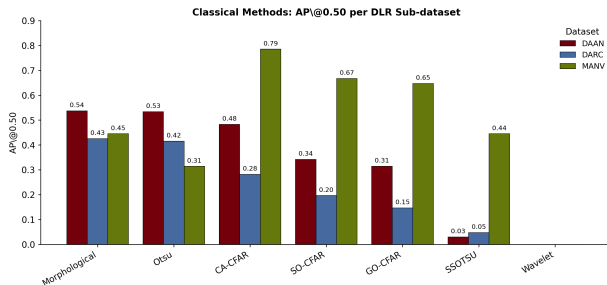
Ranked by AP@0.50. DL columns show mean $\pm 1\sigma$ over 3 seeds.

How the ranking changes across metrics



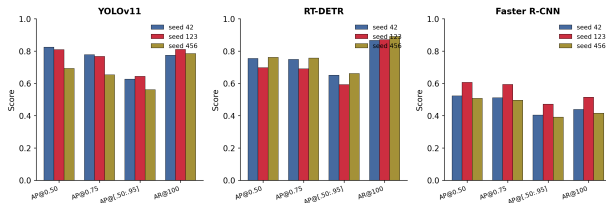
- **YOLOv11** leads AP at every IoU threshold.
- **RT-DETR** wins AR@100 — large prediction volume recovers nearly every target.
- **Morphological** and **Otsu** top the classical sub-ranking on precision metrics.
- **CA-CFAR** leads classical AR@100: CFAR is fundamentally a strong *recaller*.

Per-dataset inversion: resolution chooses the detector



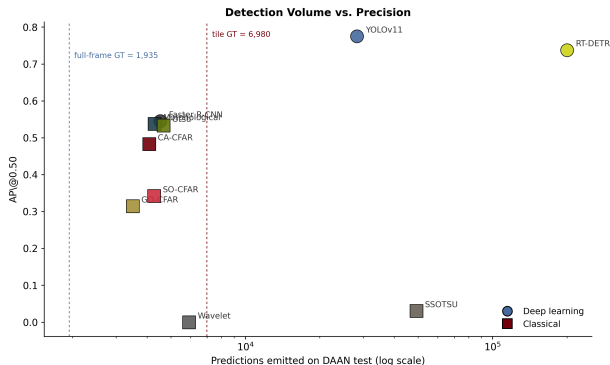
- On the close-range **MANV** (6 m/px, harbour), **CA-CFAR dominates at AP@0.50 = 0.785**.
Dense clutter, high-SNR targets, sub-metre cells — exactly where CFAR was designed to live.
- On long-range **DAAN / DARC** (12 m/px, open water), **morphological / Otsu lead at 0.53–0.54**.
- Actionable: *pick the classical detector to match the radar's resolution regime.*

Deep-learning seed stability



- All three DL detectors: $\sigma \leq 0.06$ on every metric.
- **YOLOv11** is the most consistent ($\sigma \approx 0.01$ – 0.02 on AP).
- RT-DETR next; Faster R-CNN most variable.
- No seed dominates any single metric $\rightarrow$ the training regime has converged.

Detection volume vs. precision



- **Faster R-CNN** — 4.5 k preds across 4.56 k tiles, *one-to-one with tile count*, AP = 0.55.
- Classical cluster: 4–8 k preds, AP = 0.31–0.55.
- **RT-DETR** (200 k) and **YOLOv11** (28 k) over-predict heavily, but confidence ranking preserves AP.
- **SSOTSU** (49 k / AP 0.03) and **Wavelet** are clear outliers — both broken at the scoring step.

- **Deep detectors win the benchmark, but the gap is narrower than the literature implies.**

Best classical (morphological, 0.537) is only *0.008* below Faster R-CNN and reaches *69%* of YOLOv11 with no training.

- **The classical paradigm chooses the detector** at the radar's resolution regime — CA-CFAR for 6 m/px, morphological/Otsu for 12 m/px.

- **Why our numbers sit below published results** (e.g. YOLOv11 0.776 vs reported 0.95): prior work matches against DBSCAN-padded 33×33 boxes. We use *tight* AIS-anchored GT — harder, but operationally predictive.

Known failures

- **Wavelet-SWT**: 44×47 px blobs vs GT $15 \times 15 \rightarrow AP = 0$.
- **SSOTSU**: 49 k over-predictions $\rightarrow AP = 0.03$.
- **U-Net**: training did not finish in time for this draft.
- **OS-CFAR**: coded, not yet reported.

Pending proposal deliverables

- TIDE error decomposition (classification / localization / background / duplicate / missed).
- Pairwise bootstrap significance (10 000 resamples, Bonferroni).
- Accuracy-vs-latency Pareto (FLOPs, inference time, parameter count).
- Per-size AP stratification (APs / APm / API).
- YOLOv11-OBb + SAHI re-run.

Conclusion

- First controlled cross-paradigm benchmark of **7 classical + 4 deep** detectors on X-band marine radar PPI.
- Two systematic flaws in the public DLR release corrected: **17-px polar-origin bias** and **2× MANV range-scale error**.
- Hybrid AIS-anchored / image-grounded GT generator released.
- Under the cleaner evaluation: YOLOv11 leads at **0.776 ± 0.06** , RT-DETR 0.738, Faster R-CNN 0.545; classical competitive but below, with **CA-CFAR 0.785 on MANV**.

All code, configs, checkpoints, calibration, and GT generator released.
github.com/j-vaught/marine-radar-detection